

3. StoreBackend

StoreBackend = the agent's long-term memory. = a cloud-drive, permanent, and any thread/session/user can pull from it later, no matter where the agent is running.

Simpler term, it is like a database that the agent can write files into, that does not disappear when the conversation ends.

Why:

- If the agent learns "this user prefers short answers" or "this user's project use python", we want this information to be remembered for days not just for the rest of today's chat.

```
from deepagents import create_deep_agent
from deepagents.backends import StoreBackend
from langgraph.store.memory import InMemoryStore

store = InMemoryStore() # swap for PostgresStore etc. in production

agent = create_deep_agent(
    backend=StoreBackend(namespace=lambda rt: (rt.context.user_id,)),
    store=store,
)
```

- **store** = where the data actually lives (it can be swapped for Postgres/Redis in real production)
- **namespace** = whose memory this is. If the names not separately specified here, everyone's memories get mixed together. So this part is a must for the app with multiple users.

Comparison

Backend	Lives in	Cross-thread?	Survives restart?	Real file on disk?
StateBackend	LangGraph state	✗	✗	✗
FilesystemBackend	Your disk	✓	✓	✓
StoreBackend	A LangGraph store	✓	only w/ persistent store	✗

-